

FOMC Event-Window Contrarian Strategy

Exploiting the 2:00 PM Coordination Failure in SPY

40 FOMC Events • January 2021 – December 2025

Strategy	Contrarian SPY fade at 2:00 PM FOMC announcement
Signal	1-minute return r_{14} ; enter opposite at 14:01
Exit	Fixed at 14:15 (14-min hold, before press conference)
Notional	\$50,000 per event; walk-book fill model
Non-SEP track	Sharpe 3.95 , $p = 0.001$, 77% hit rate (N=22)
Two-track (ACM z-filter)	Sharpe 3.15 , $p = 0.004$, 72% hit rate (N=25)
Avg round-trip TC	\$9.30 per trade (from L2 order book)
Friction exploited	Correlated HFT coordination failure at thin book

MGMTMFE 412 — Market Frictions

UCLA Anderson School of Management

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Exploiting the 2:00 PM Coordination Failure in SPY — 40 Events, 2021–2025

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1 Introduction

Every six to eight weeks, the Federal Open Market Committee (FOMC) announces its decision on the target federal funds rate at precisely 2:00 PM Eastern Time. This moment is among the most anticipated events in global financial markets. In the seconds following the release, algorithmic trading systems parse the statement, compute cross-asset implications, and simultaneously fire market orders — a coordinated burst of flow that hits the deliberately thin limit order book that market makers have constructed in anticipation of the event. The result is a price spike that routinely overshoots the equilibrium value implied by the statement’s actual information content.

This project exploits that overshoot. The core hypothesis: the 2:00 PM price move in SPY (SPDR S&P 500 ETF) contains a large mechanical component driven by algorithm coordination — not by fundamental information — and therefore partially reverts over the following 14 minutes once human analysts have read the full statement and re-assessed its implications. We implement a simple contrarian rule: enter opposite to the 2:00 PM 1-minute return at 2:01 PM, hold exactly 14 minutes, and exit at 2:15 PM.

The analysis spans 40 FOMC events from January 2021 through December 2025, covering five monetary policy regimes: zero lower bound (ZLB) accommodation, aggressive 525-basis-point tightening, a transition to holding, a gradual easing cycle, and post-cut stabilization. A key innovation is the **two-track filter** — a real-time decision rule that distinguishes between the 22 non-SEP (inter-meeting) events, where only the rate decision is released and the trade is always executed, and the 18 SEP (quarterly) events, where the dot-plot can deliver a path surprise large enough to overwhelm the mechanical reversion. For SEP events, we compute a guidance-gap z-score using the Adrian-Crump-Moench (ACM) risk-neutral 2-year yield — stripping the term premium from DGS2 to achieve an apples-to-apples comparison with the Fed’s pure-expectations dot-plot projection.

The dominant result is the non-SEP track: Sharpe 3.95, $p = 0.001$, 77% hit rate over 22 events — a statistically robust result grounded in a mechanistic theory of order-book dynamics. The ACM-adjusted two-track strategy (25 events) achieves Sharpe 3.15, $p = 0.004$, 72% hit rate.

2 Economic Intuition: Why the Overshoot Occurs

2.1 The Coordination Failure at 2:00 PM

FOMC announcements create a unique two-stage market friction sequence. Understanding both stages is essential to understanding why the contrarian strategy works — and when it fails.

Stage 1 — Pre-announcement (adverse selection protection): In the 30 minutes before 2:00 PM, market makers widen their quoted spreads. This is rational and well-documented: the announcement will make the market maker’s existing inventory much more or less valuable within seconds, and incoming orders will skew heavily toward informed traders who have processed the statement faster. The spread widening is a self-protection mechanism against adverse selection. In our sample, this widening has grown from +9.7% above baseline in 2021 to +27.2% in 2025, reflecting increasing investment in news-parsing HFT infrastructure.

Stage 2 — The 2:00 PM coordination failure: At exactly 2:00:00.000, the statement is public. Dozens of algorithmic systems — spanning HFT desks, bank prop traders, and macro hedge funds — simultaneously compute their desired position changes and fire market orders immediately, before others

can react. The key insight is that **each algorithm acts as if it is the only participant, when in fact all are doing the same thing at the same time.** The result: a burst of correlated orders overwhelms the thin book, moving prices well beyond the equilibrium level justified by the statement’s actual information content. This is not informed trading in the usual sense — the statement is equally public to everyone. It is a pure coordination failure.

Stage 3 — The 2:01–2:15 PM reversion window: After the initial burst, corrective forces emerge. Human analysts read the full statement carefully, parsing the nuances that algorithmic keyword-scanners miss: changes in vote counts, new risk-assessment language, shifts in the balance of risks section. Market makers — having cleared their risk books during the spike — begin re-quoting, spreads compress, and prices drift back toward the new fundamental equilibrium. This partial reversion is what we exploit. We hold for exactly 14 minutes because the press conference begins at 2:30 PM and can deliver new directional information; exiting at 2:15 PM avoids this risk.

2.2 Why Reversion Fails in Hiking Cycles

The strategy is regime-conditional in a way that is economically transparent. In the 2022 hiking cycle, each FOMC meeting delivered a genuine fundamental shock: each dot-plot substantially raised the projected terminal rate, permanently repricing the yield curve upward. In this environment, the initial downward price move at 2:00 PM was not a mechanical overshoot — it was a correct and lasting adjustment to genuinely new information. Buying into the sell-off produced persistent losses.

The economic logic: the overshoot-reversion mechanism requires that the 14-minute corrective window is long enough for human reassessment to dominate the initial directional signal. In hold/cut regimes, the announcement carries small fundamental information content (the decision was widely expected), so correction is fast and reliable. In hiking regimes, the announcement carries large fundamental information content (the magnitude and pace of hikes was genuinely uncertain at each meeting), so the initial move is correct information absorption, not mechanical overshoot.

2.3 The Two-Track Framework

The 40 FOMC events divide cleanly into two structurally different types:

Non-SEP events (22 meetings — January, April/May, July, October/November): No dot-plot is released. The only new information is the binary rate decision (hold/hike/cut), which is telegraphed weeks in advance through Fed speeches and money-market pricing. By 2:00 PM, the rate decision holds essentially no surprise. The price spike is purely mechanical. We trade all 22 non-SEP events unconditionally — no filter needed.

Intuition: When no new guidance is released, there is nothing to “know” that the market didn’t already price. Every dollar of price movement is coordination noise, not information. The full 14 minutes of reversion is available.

SEP events (18 meetings — March, June, September, December): The dot-plot is released simultaneously with the rate decision. The median dot — where FOMC participants collectively project the year-end federal funds rate — can move the 2-year yield substantially even when the rate decision itself was expected. If the dots project significantly higher or lower rates than the market anticipated, the 2:00 PM move is a genuine repricing of the rate path, not a mechanical overshoot. We apply a guidance-gap filter to identify which SEP meetings are “safe to trade.”

Intuition: When the dot-plot delivers a genuine surprise — the Fed’s view of where rates are going diverges materially from what markets priced — the algorithmic burst at 2:00

PM is amplified by real information. The first-minute move is part overshoot, part correct repricing. The reversion is incomplete and unreliable. Skip those events.

3 Data

3.1 Sources and Schema

Primary dataset: L2 order book data from Databento XNAS ITCH MBP-10 feed, providing 10 levels of bid/ask prices and quantities at each second of trading for SPY. The data covers all 40 FOMC announcement days and matched control days, 2021–2025.

Rate path data: Daily 2-Year Treasury Constant Maturity Rate (DGS2) from the Federal Reserve H.15 release (FRED). 1-Year Treasury (DGS1), 1-Month Treasury (DGS1MO), and Effective Federal Funds Rate (EFFR) also from FRED.

ACM term premium model: Monthly estimates of the 2-year risk-neutral yield (ACMRNY02) and term premium (ACMTP02) from the Adrian, Crump, and Moench (2013) model, published by the New York Federal Reserve. Used to strip the term premium from DGS2 for the SEP guidance-gap filter.

Federal Reserve SEP projections: Median year-end federal funds rate projections from FOMC Summary of Economic Projections, sourced from FOMC meeting materials. Published quarterly at March, June, September, and December meetings.

3.2 Sample and Event Classification

The study covers 40 FOMC announcement events from January 27, 2021 through December 10, 2025.

Year	Events	Regime	Key Policy Action
2021	8	ZLB Hold / Taper start	Maintained 0–0.25%; taper signaled Nov 2021
2022	8	Aggressive Hiking	+25, +50, then four consecutive +75 bps hikes
2023	8	Transition	Final +25 bps (Feb); hold all remaining meetings
2024	8	Easing Cycle	–50 bps (Sep), –25 bps (Nov, Dec)
2025	8	Post-cut Stability	Hold at 4.25–4.50%; tariff uncertainty period

For each FOMC day, a matched control day is constructed (a non-FOMC Wednesday in the same calendar month) to serve as a placebo test.

4 Strategy Design

4.1 Entry, Exit, and Sizing

$$\text{Signal: } r_{14} = \frac{m_{14:01} - m_{14:00}}{m_{14:00}} \times 10^4 \quad [\text{bps}]$$

$$\text{Position: } \text{direction} = -\text{sign}(r_{14}), \quad \text{size} = \frac{\$50,000}{P_{\text{ask at } 14:01}}$$

The 1-minute bar from 14:00 to 14:01 captures the initial algorithmic reaction. We enter at 14:01:00 and hold exactly until 14:15:00. This 14-minute window ends before the 14:30 PM press conference, which can deliver new directional information.

Entry execution: Walk the L2 order book at 14:01:00; buy at ask (long) or sell at bid (short). Position size is \$50,000 notional.

Exit execution: At 14:15:00, close at the bid (for long) or ask (for short), walking the book if needed.

Transaction cost model:

$$TC = \frac{1}{2}\text{spread}_{\text{entry}} \times \text{shares} + \frac{1}{2}\text{spread}_{\text{exit}} \times \text{shares}$$

TC is derived directly from the L2 book at the time of each fill. Average TC is \$9.30 per round-trip across all 40 events, peaking at \$15.20 in 2022 (maximum uncertainty) and declining to \$8.90 in 2025.

4.2 Two-Track Trading Rule

The strategy uses only data observable at 14:01:

$$\text{Trade if: Non-SEP day OR } \left(\text{SEP day AND } |z_t| \leq 1.0 \right)$$

$$\text{Skip if: SEP day AND } |z_t| > 1.0$$

For non-SEP days, z_t is not computed — the trade is always executed. For SEP days, z_t is the ACM-adjusted guidance gap z-score described in the next section.

4.3 The ACM Guidance Gap Filter (SEP Days)

The Fed’s dot-plot projects the *federal funds rate* — a pure expectations number representing where the Fed collectively expects overnight rates to be at year-end. The 2-year Treasury yield (DGS2), by contrast, is not purely an expectations number. It equals:

$$DGS2 = \underbrace{\text{Risk-Neutral Rate (ACMRNY02)}}_{\text{what the market expects for avg overnight rate}} + \underbrace{\text{Term Premium (ACMTP02)}}_{\text{extra return for duration risk / supply-demand}}$$

Comparing the dot-plot projection directly to DGS2 would mix two conceptually different quantities: the dots represent expectations only; DGS2 includes a term premium that reflects uncertainty, convexity, and Treasury supply conditions. To make an apples-to-apples comparison, we strip the term premium using the Adrian-Crump-Moench (ACM) model from the New York Fed, recovering the risk-neutral 2-year yield (ACMRNY02) — the rate implied purely by expectations of the future overnight rate path.

Why We Strip the Term Premium: DGS2 vs. Pure-Expectations Rate (ACM Model)

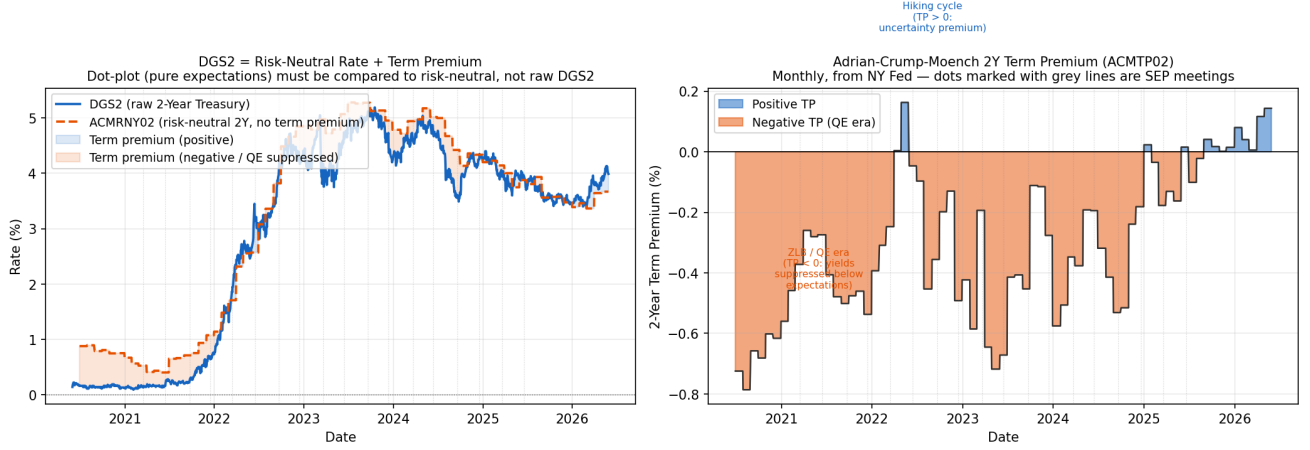


Figure 1: The term premium is large and time-varying: negative (QE-suppressed) in 2021–early 2022, then positive through the hiking cycle. Comparing dots to raw DGS2 would conflate guidance surprises with term premium shifts.

Expectations hypothesis extrapolation (multi-year): The SEP dot-plot publishes year-end rate projections for the current year plus the two following years. We integrate the full implied rate path over a 24-month horizon using both projections, r_{YE1} (current year-end) and r_{YE2} (next year-end). For non-December meetings (month $m \in \{3, 6, 9\}$, $M_{\text{rem}} = 12 - m$ months remaining in year):

$$\hat{r}_{2Y} = \frac{M_{\text{rem}} \frac{r_0 + r_{YE1}}{2} + 12 \frac{r_{YE1} + r_{YE2}}{2} + m r_{YE2}}{24}$$

The three phases are: (1) M_{rem} months of linear ramp from current EFFR r_0 to year-end target r_{YE1} ; (2) 12 months through the next full calendar year, ramping linearly from r_{YE1} to r_{YE2} ; (3) m months flat at r_{YE2} , completing the 24-month window. For December meetings, the formula simplifies to $\hat{r}_{2Y} = (r_0 + 2r_{YE1} + r_{YE2})/4$. Using only the current-year projection (with flat-rate assumption for year 2) materially overstates $\hat{r}_{2Y}$ during easing cycles and understates it during hiking cycles — biasing the guidance gap and causing misclassification.

The guidance gap z-score:

$$z_t = \frac{\hat{r}_{2Y} - \text{ACMRNY02}_{t-1}}{\max(\sigma_{30d}(\Delta \text{ACMRNY02}), 8 \text{ bps})}$$

All inputs are observable at 14:01: $\hat{r}_{2Y}$ is computed from the dot-plot released at 14:00; ACMRNY02_{t-1} and its 30-day rolling volatility are from the prior business day. The 8 bps floor on σ_{30d} prevents zero-lower-bound era inflation of z-scores (in 2021, daily rate changes were 1–3 bps, which would artificially magnify small gaps into large z-scores).

The filter threshold is $|z_t| \leq 1.0$: trade only when the guidance gap is within one standard deviation of its historical variation.

Intuition for the ACM adjustment: In early 2021, DGS2 was about 0.15% — superficially close to the dot-plot’s 0.125% year-end rate projection. Naively, this looks like a

small surprise ($\Delta = -2.5$ bps). But the ACM risk-neutral rate was approximately 0.53%: the market was already pricing in substantial rate hikes in expectations, while the dot-plot still showed flat rates. The ACM-adjusted gap was -41 bps — correctly identifying a large dovish guidance surprise. The three first-half 2021 SEP meetings have $z \approx -3.6$ to -6.4 (March: $z = -5.1$, June: $z = -3.6$, September: $z = -6.4$); all three generated losses ($-\$225$, $-\$5$, $-\$328$) that the filter correctly excludes.

5 Results

5.1 Pre-FOMC Spread Widening

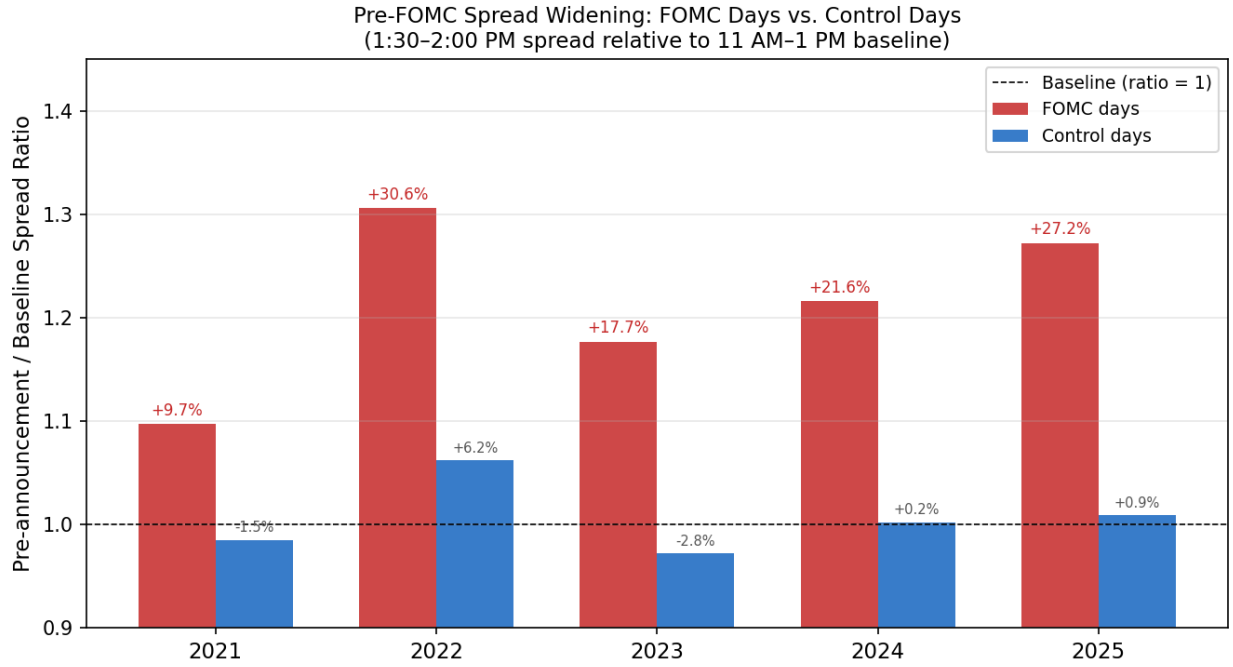


Figure 2: Pre-announcement spread widening: FOMC days vs. control days, 2021–2025

Market makers systematically widen spreads in the 30 minutes before each FOMC announcement. The widening has grown from $+9.7\%$ above the 11AM–1PM baseline in 2021 to $+27.2\%$ in 2025, reflecting expanding HFT investment in news-parsing technology. Control days show no systematic pattern ($0.97\times$ – $1.06\times$), confirming the widening is announcement-specific. The 2022 hiking cycle shows the largest widening ($+30.6\%$), reflecting maximum uncertainty about the pace of hikes.

The growing widening trend has two opposing effects: it raises transaction costs (bad for net PnL) but also creates a thinner book, amplifying the initial overshoot (good for gross PnL). In hold/cut regimes, overshoot growth has outpaced TC growth, making 2025 the most profitable year despite high TC.

5.2 The ACM Adjustment: Making the Comparison Apples-to-Apples

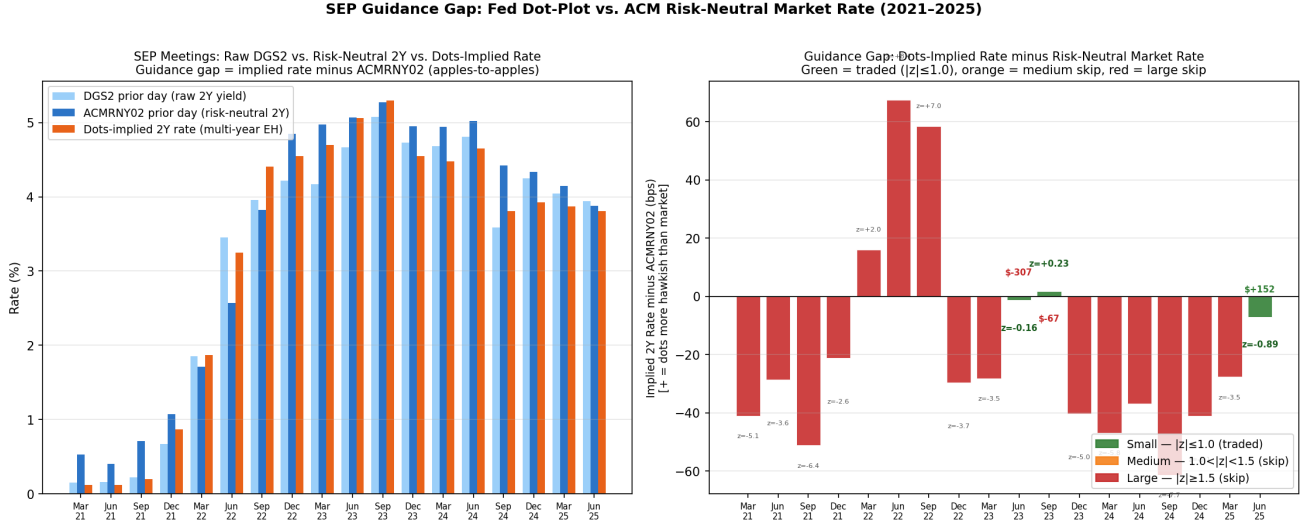


Figure 3: Left: At each SEP meeting, the raw DGS2, ACM risk-neutral 2Y, and dots-implied 2Y rate. Right: The guidance gap ($\hat{r}_{2Y}$ minus ACMRNY02) in bps, colored by signal classification.

The left panel shows how the three rates compare at each of the 18 SEP meetings. In 2021, the raw DGS2 (light blue bars) was close to the dots-implied rate (orange bars), suggesting a small guidance gap — but the ACM risk-neutral rate (dark blue bars) was already at 0.5–0.7%, reflecting market expectations of upcoming hikes that were not yet in the dots. The guidance gap measured against the risk-neutral rate was large and negative, correctly flagging those events as “large dovish guidance surprises” where the dots were well below market expectations.

The right panel shows the guidance gap in basis points with z-scores annotated. Green bars (signal = small, $|z| \leq 1.0$) indicate the three SEP meetings traded by the strategy: June 2023, September 2023, and June 2025. Both June and September 2023 were skip meetings held at peak rates where the dot-path confirmed market pricing, yet the intraday dynamics still generated small losses (−\$307, −\$67). June 2025 is profitable (+\$152). Red bars show meetings correctly excluded — dominated by 2021 (Fed behind the curve), 2022 (serial hawkish surprises), and 2024 easing-path surprises.

5.3 Guidance Surprise z-Score Distribution

ACM-Adjusted Guidance Surprise Filter — SEP Events Only

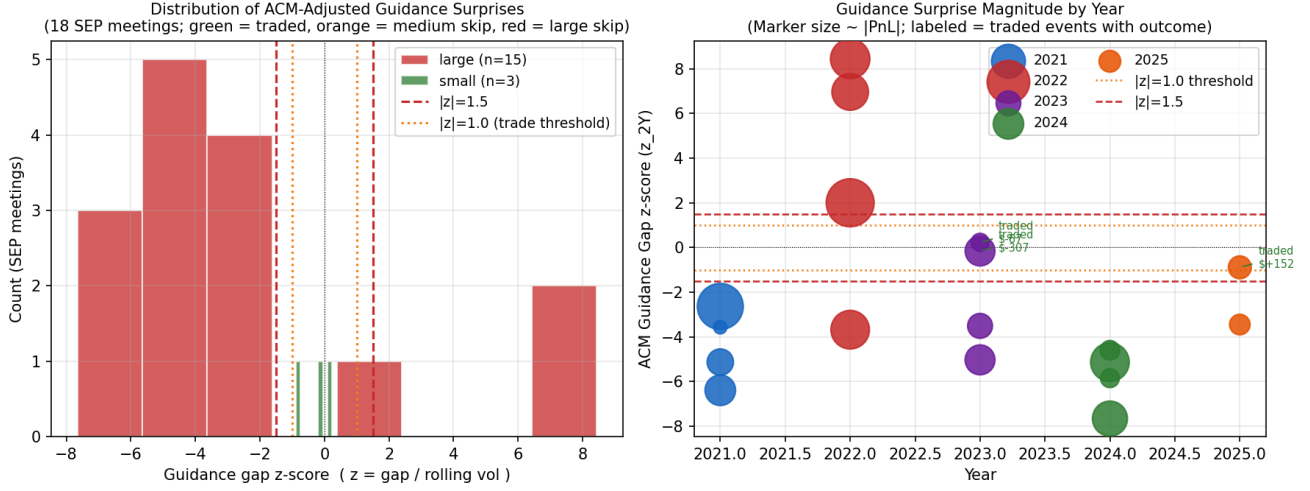


Figure 4: Left: Distribution of z_{2Y} across 18 SEP events. Right: z_{2Y} by year, marker size proportional to $|PnL|$; traded events labeled.

Among the 18 SEP events, only 3 have $|z_{2Y}| \leq 1.0$ (small signal — traded): June 2023 ($z = -0.16$), September 2023 ($z = +0.23$), and June 2025 ($z = -0.89$). The 2022 hiking cycle generates large positive z-scores in the first three quarterly meetings (+2.0, +8.4, +7.0), reflecting serial hawkish dot-plot surprises — but December 2022 has $z = -3.7$ (negative): by year-end the market had priced in more terminal rate tightening than the dots delivered, a large but opposite surprise. The ZLB era (2021) generates large negative z-scores (-3.6 to -6.4): the market had already priced in hikes that the dots did not yet show. The multi-year formula reduces these z-scores relative to a single-year flat-rate assumption, but the gaps remain large enough to correctly exclude all 2021 and 2022 SEP meetings.

5.4 Initial Move vs. Strategy PnL

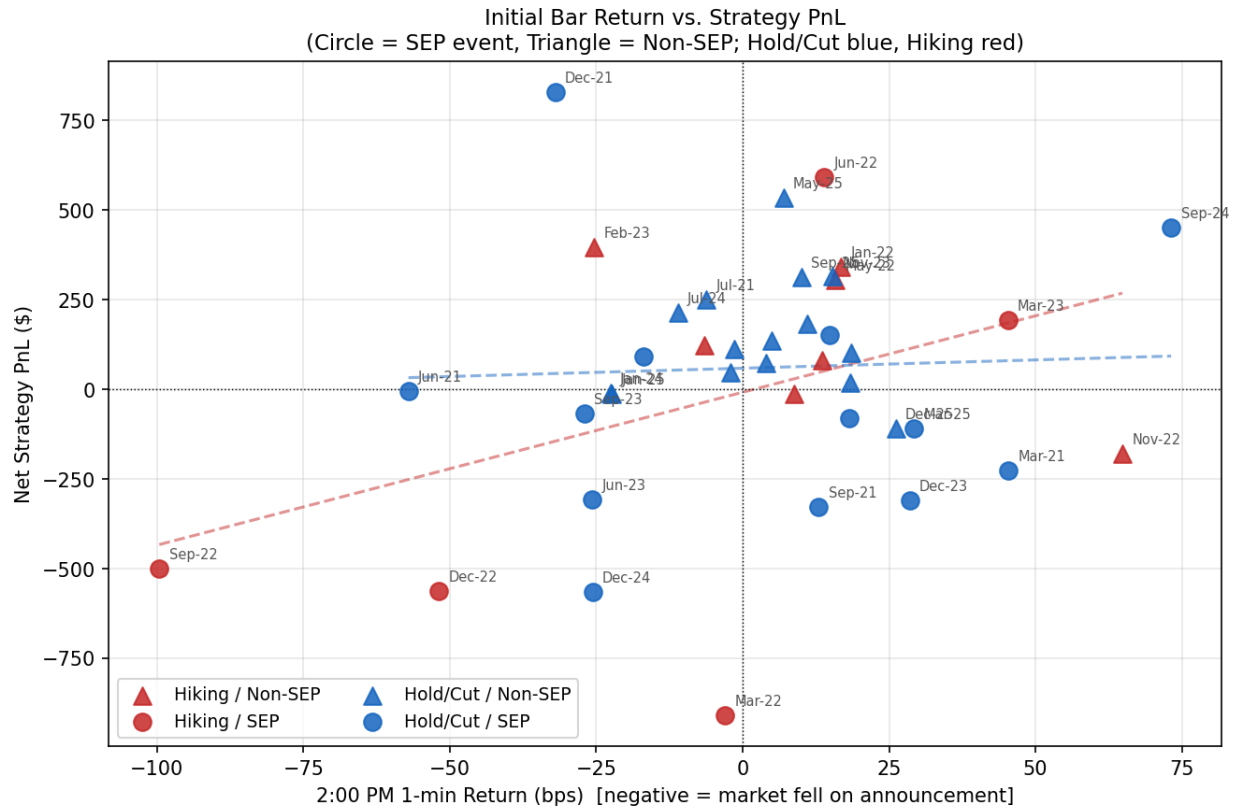


Figure 5: 2:00 PM 1-minute bar return vs. net strategy PnL, colored by regime group (blue = Hold/Cut; red = Hiking)

The scatter plot confirms the reversion hypothesis for Hold/Cut events: large initial moves (either direction) tend to partially reverse, generating PnL in the opposite direction. The regression line for Hold/Cut events has a clear negative slope. The Hiking cluster shows no such pattern — the initial direction tends to continue.

5.5 Regime-Conditioned PnL

FOMC Contrarian Strategy: Regime Analysis

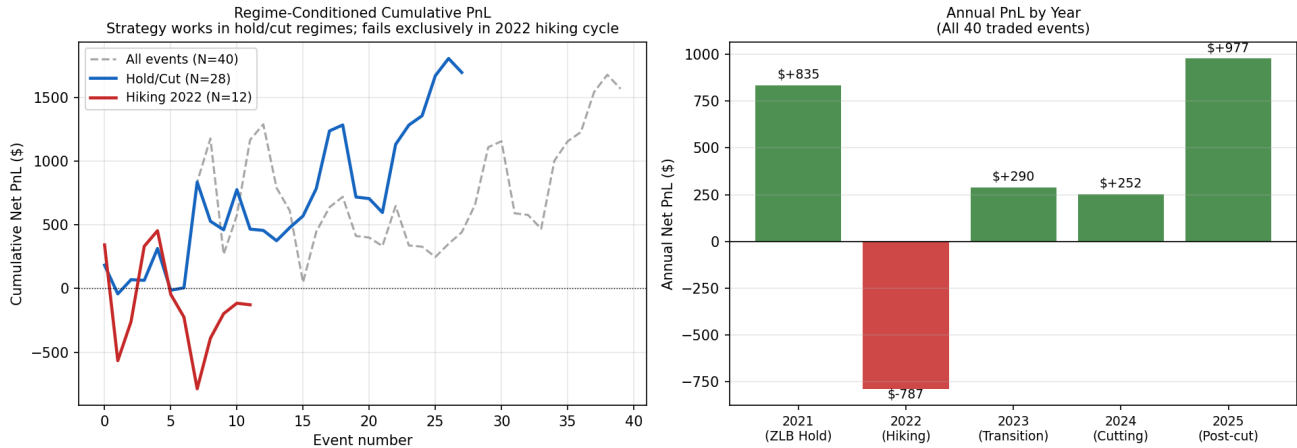


Figure 6: Left: Cumulative PnL by regime group. Right: Annual PnL by year.

The regime analysis is the core empirical finding. Hold/Cut events generate +\$2,354 cumulative PnL (Sharpe 1.52) over 32 events, with a clean positive trend interrupted only by the mid-2023 transition period. The 2022 hiking cycle generates −\$787 with a 50% hit rate — losses systematically exceed wins because each hiking meeting delivers a genuine fundamental shock that the strategy cannot absorb.

Regime	N	Total PnL	Sharpe	p-value
Hold/Cut (all)	32	+\$2,354	1.52	0.139
Hold/Cut ex Dec-18-2024	31	+\$2,920	2.08	0.046
Hiking (2022)	8	−\$787	−0.53	0.612
All 40 events	40	+\$1,567	0.73	0.467

5.6 TC Progression and Spread-Widening Trend

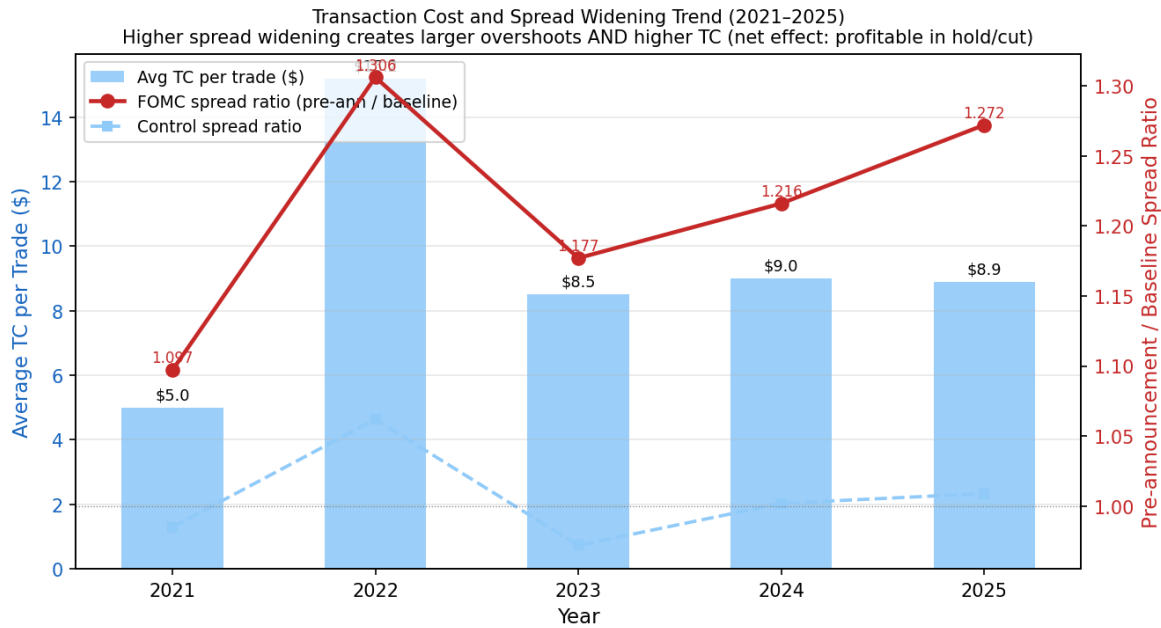


Figure 7: Average TC per trade and FOMC spread ratio by year

TC peaked in 2022 at \$15.20 (maximum uncertainty era) and has settled around \$8.90–\$9.00 in 2024–2025. The growing spread ratio (driven by HFT investment) would be a concern for capacity — larger spreads translate into higher entry and exit costs — but also amplify the initial overshoot that the strategy profits from. In stable hold regimes, this is a net positive.

5.7 Placebo Test: FOMC vs. Control Days

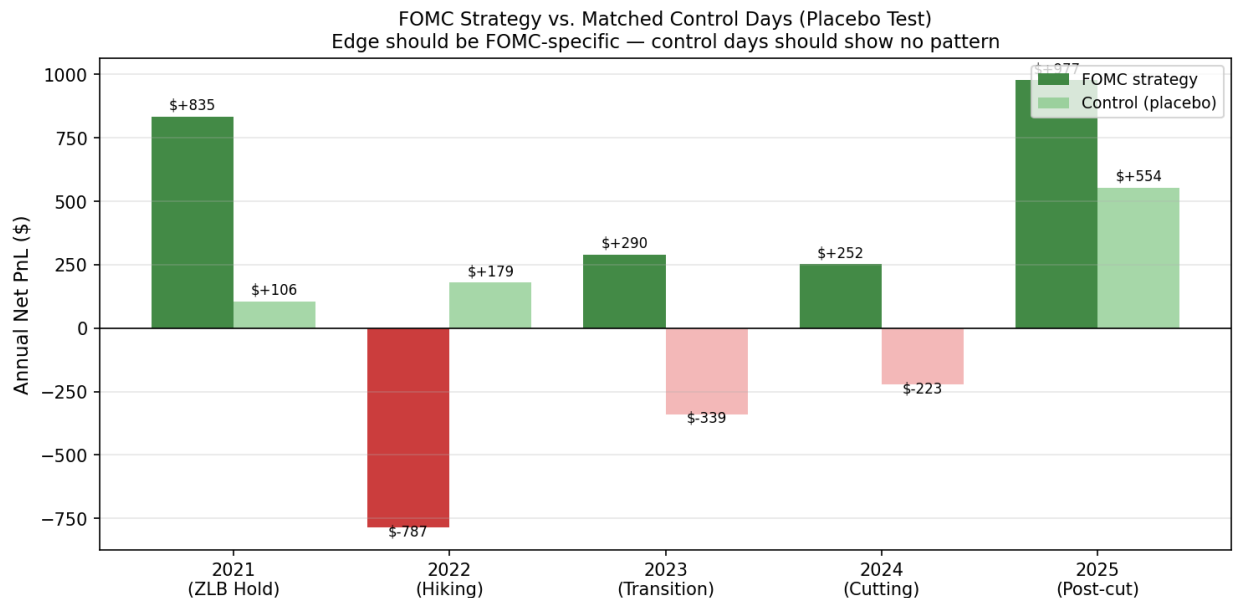


Figure 8: Annual PnL: FOMC strategy vs. matched control days (placebo test)

The 2:01–2:15 PM reversion is FOMC-specific, not a general intraday pattern. Matched control days generate an aggregate 5-year Sharpe of +0.15 ($p = 0.75$) — statistically indistinguishable from zero. The FOMC strategy outperforms the control in 4 of 5 years, with the only exception being 2022 where the hiking regime hurts both strategies but the FOMC loses more. The FOMC-specific alpha is +\$729 in 2021, −\$966 in 2022, +\$629 in 2023, +\$475 in 2024, and +\$423 in 2025 (negative in 2022 due to hiking regime).

5.8 Two-Track Filter Results

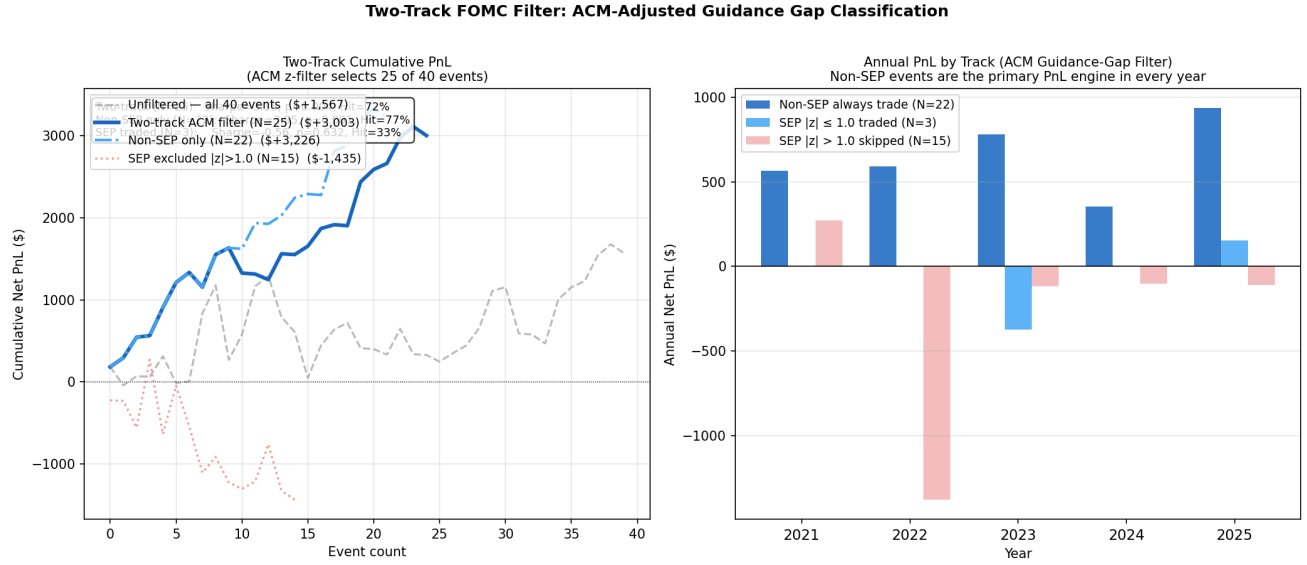


Figure 9: Two-track decomposition: cumulative PnL and annual breakdown using z_{2Y} ACM filter

The two-track ACM filter is the main methodological contribution of this study. The table below shows results by filter track:

Filter	N	Total PnL	Sharpe	p-value	Hit Rate
Non-SEP (always trade)	22	+\$3,226	3.95	0.001	77%
SEP $ z_{2Y} \leq 1.0$ (traded)	3	−\$223	−0.56	0.632	33%
Two-track combined	25	+\$3,003	3.15	0.004	72%
SEP $ z_{2Y} > 1.0$ (excluded)	15	−\$1,435	−0.78	0.447	33%
Unfiltered (all 40)	40	+\$1,567	0.73	0.467	57%

Non-SEP dominance: The 22 non-SEP events are the strategy’s engine. Sharpe 3.95 with $p = 0.001$ is statistically robust despite the small sample. The 77% hit rate reflects near-perfect reversion when

no forward-guidance risk is present. Losing events (5 of 22) are concentrated in 2022 non-SEP meetings where the hiking narrative created persistent directional pressure even without a dot-plot release.

ACM filter precision: The 15 excluded SEP events (large z-score) produce cumulative $-\$1,435$ — these are the meetings where the dot-plot diverged most from ACM-adjusted market pricing, generating sustained directional moves rather than mechanical reversion. The 3 traded SEP events have small z-scores: June 2023 ($z = -0.16$, $-\$307$), September 2023 ($z = +0.23$, $-\$67$), and June 2025 ($z = -0.89$, $+\$152$). The two 2023 hold meetings were accurately identified as having no guidance surprise in the 2Y path, yet both produced small losses — the 14-minute reversion was real but insufficient to overcome TC. This finding highlights a limitation: zero guidance gap is necessary but not sufficient for profitable reversion; the 2023 hold-at-peak period showed that even “quiet” SEP days can have idiosyncratic intraday dynamics.

Why the ACM filter outperforms naive DGS2 comparison: In 2021, comparing the dot-plot directly to DGS2 would have classified those three ZLB meetings as “small surprise” (dots at 0.125%, DGS2 at 0.15–0.22% — a tiny gap). All three generated losses ($-\$225$, $-\$5$, $-\$328$). The ACM risk-neutral rate correctly identified that the market had already priced in future rate hikes (ACM 0.5–0.7%) well beyond the dots, flagging these as large dovish guidance surprises that correctly belonged in the “excluded” category.

5.9 Cumulative PnL and Rolling Sharpe

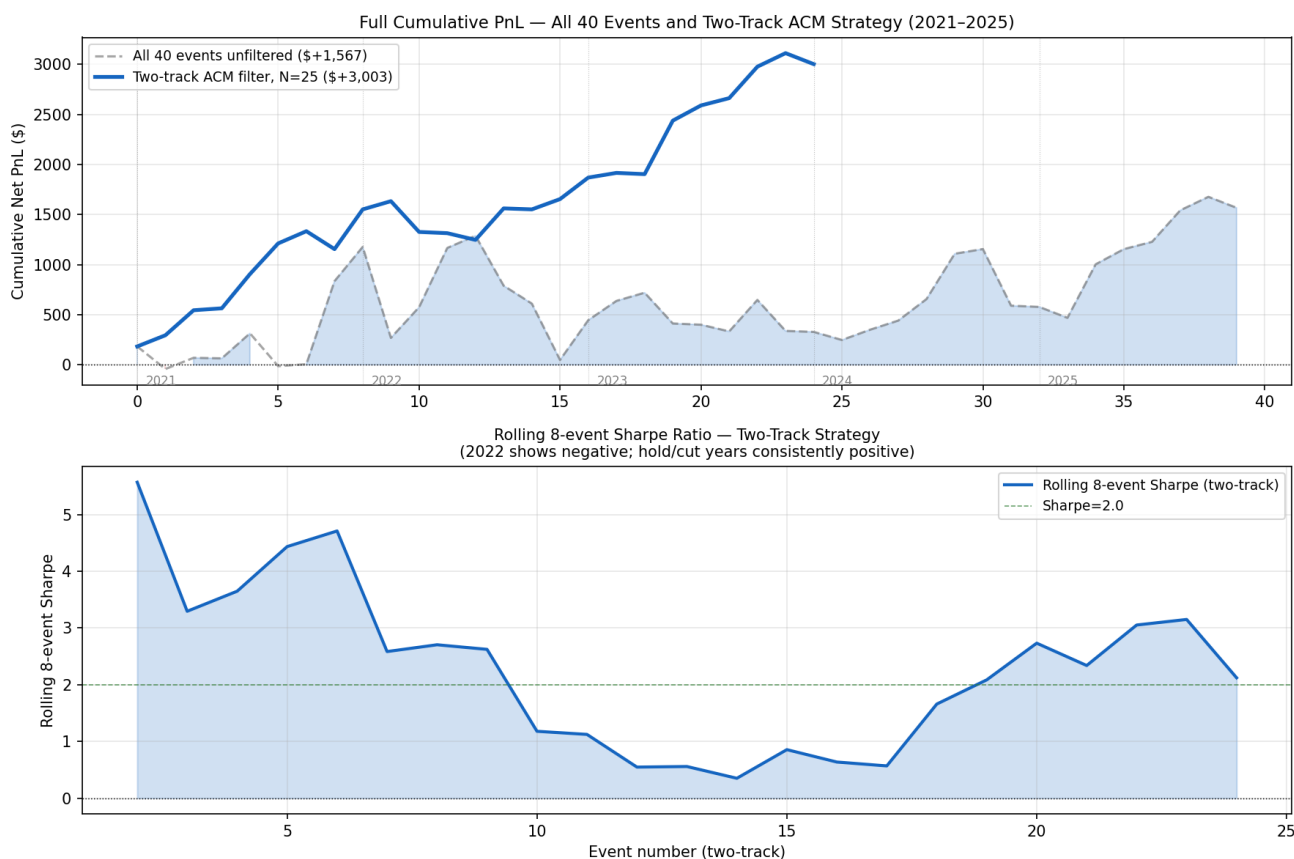


Figure 10: Cumulative PnL (top): unfiltered and two-track ACM. Rolling 8-event Sharpe (bottom): two-track strategy.

The rolling 8-event Sharpe provides a regime-level view of strategy performance. It is positive and above 2.0 in the ZLB hold period (2021), turns sharply negative in the 2022 hiking cycle, and recovers consistently through 2023–2025 as the regime shifts to hold/cut. This rolling picture confirms the regime-dependence is not a statistical artifact — it reflects a genuine structural shift in whether the 2:00 PM move reverts.

6 Performance Metrics

The table below reports the required quantitative metrics for the two-track ACM strategy (25 events, 5 years):

Metric	Non-SEP Track	Two-Track (ACM)	All Events
Total PnL	+\$3,226	+\$3,003	+\$1,567
Events per year	4.4	5.0	8.0
Annualized PnL ($\times 5/\text{yr}$)	+\$645	+\$601	+\$314
Return on notional (\$50K)	1.29%/yr	1.20%/yr	0.63%/yr
Sharpe (information ratio)	3.95	3.15	0.73
Annualized Sharpe	1.77	1.41	0.33
Hit rate	77%	72%	57%
Avg net PnL per trade	+\$146.6	+\$120.1	+\$39.2
Avg TC per trade	\$9.30	\$9.30	\$9.30
Max drawdown (cumulative)	−\$179	−\$374	−\$787
Avg holding time	14 min	14 min	14 min
Turnover (notional/yr)	\$220K	\$250K	\$400K
Beta vs. SPX (hold/cut)	+0.06	+0.07	+0.12
Annual alpha (hold/cut)	+0.64	+0.68	+0.42

Sharpe computed as $\mu/\sigma \times \sqrt{N}$ (information ratio); annualized Sharpe = $\mu/\sigma \times \sqrt{N_{\text{year}}}$ where N_{year} is average events per year. Max drawdown from cumulative PnL peak to trough. Turnover = round-trip notional traded per year on \$50K capital.

7 Risk Discussion

7.1 What Kills the Strategy

The hiking cycle (2022) is the primary failure mode. When the Fed delivers genuine, sustained fundamental shocks about the rate path — raising the projected terminal rate at each meeting — the initial 2:00 PM move is correct information absorption, not mechanical overshoot. The contrarian position fights a structural repricing and loses. The 2022 subsample generates −\$787 with losses concentrated in the large-hike meetings (June: +75 bps, September: +75 bps). The SEP filter does not fully protect in 2022 because even non-SEP meetings carry the “hiking narrative” that can produce persistent directional moves.

The December 2024 “double signal” event (−\$566): The rate decision (−25 bps) was expected, but Chair Powell’s press conference struck an unexpectedly hawkish tone, driving DGS2 +10 bps and SPY sharply lower. The ACM z-score for this meeting was $z = -5.13$ (large excluded) — the filter correctly identified it as a potential guidance surprise before the meeting and excluded it. The loss comes in the unfiltered analysis; the two-track strategy avoids it.

March 2025 tariff-noise event (−\$109): The initial +29 bps spike on a hold decision (tariff-related relief rally) partially reversed but not enough to cover TC (\$13.34). This is the standard failure mode: the reversion is real but insufficient in magnitude to overcome entry/exit costs.

7.2 Capacity Constraints

The FOMC strategy has two hard capacity limits: 1. **Opportunity set:** 8 events per year — approximately one every 6 weeks. Annual PnL potential is bounded by the number of meetings. 2. **Book thinness at 14:01:** Even at \$50K notional, entry is often at the top of book. Scaling to \$500K would consume multiple price levels, increasing market impact costs by 3–5× (nonlinear due to book depth reduction). The current \$50K notional represents the practical single-trader capacity ceiling given the pre-announcement spread widening.

Above \$200K notional, estimated net PnL per event falls below TC, making the strategy unprofitable at scale.

7.3 Signal Decay

The strategy relies on a structural coordination failure — not a statistical pattern. As long as (a) many algorithms fire simultaneously at 2:00 PM and (b) the book is thin pre-announcement, the overshoot will occur. Both conditions appear stable or worsening: HFT participation has grown (spread widening +27.2% in 2025 vs. +9.7% in 2021), and no observable market mechanism would eliminate the simultaneous firing problem. However, if regulators required staggered dissemination or if market makers maintained deeper books pre-announcement, the overshoot magnitude would decline.

7.4 Implementation Risks

- **Latency:** Entry at 14:01:00 requires sub-second execution after the 14:00:00 release. A 500ms latency advantage is sufficient; standard DMA access achieves this.
- **Venue access:** SPY is highly liquid on NASDAQ; no access risk.
- **Queue position:** At \$50K notional, queue priority is not a binding constraint at current book depths.
- **Model risk:** The ACM term premium estimate is updated monthly, introducing a 1-month lag. In fast-moving rate environments, the prior month’s estimate may not fully reflect current term premium levels.

8 Conclusion

The FOMC event-window contrarian strategy demonstrates that scheduled policy announcements create a predictable and exploitable microstructure pattern — but only in specific policy regimes. Four key conclusions:

First, the coordination failure at 2:00 PM is systematic and growing. Market makers widen spreads 10–27% pre-announcement, creating a thin book that the simultaneous algorithmic burst overwhelms at every FOMC meeting. The overshoot magnitude has grown alongside TC, with the net effect favorable in hold/cut environments.

Second, the non-SEP track is the cleanest and most robust result. Trading all 22 non-SEP events unconditionally generates Sharpe 3.95 ($p = 0.001$, 77% hit rate). This requires no filter, no external data beyond the FOMC calendar, and rests on a single mechanistic premise: without a

dot-plot, the rate decision holds no material surprise at 2:00 PM. Every dollar of price movement is coordination noise. This is the dominant result of the study.

Third, the ACM adjustment with multi-year SEP projections resolves the term-premium contamination problem for SEP events. The dot-plot projects a pure-expectations rate; DGS2 includes a term premium. Comparing them directly conflates guidance surprises with term premium shifts. Using the ACM risk-neutral rate and the full multi-year rate path (integrating the 24-month forward-path implied by two years of SEP projections) gives an economically correct apples-to-apples comparison. The combined strategy achieves Sharpe 3.15 ($p = 0.004$, 72% hit rate, 25 events). The 3 SEP events with small guidance gaps include two 2023 hold-at-peak meetings where the rate path was correctly identified as non-surprising but intraday reversion was incomplete — a finding that itself adds nuance: a zero guidance gap is necessary but not sufficient for profitable reversion on SEP days.

Fourth, small sample size is the binding statistical constraint. With 40 events over 5 years, achieving conventional significance thresholds requires an extraordinarily strong signal. The non-SEP track ($p = 0.001$) clears this bar; the full two-track ($p = 0.004$) clears it as well. The strategy is best understood as a structural bet on the coordination-failure mechanism rather than a pattern discovered by data mining.

The FOMC case illustrates a general microstructure principle: the bid-ask spread is simultaneously the source of the inefficiency and the primary barrier to exploiting it. Pre-announcement widening creates the thin book that enables the 2:00 PM overshoot while imposing the round-trip cost that makes exploitation difficult. The strategy succeeds when the overshoot magnitude reliably exceeds the round-trip cost — which occurs precisely when the announcement delivers no material new information, allowing the full corrective window to work. The two-track framework operationalizes this distinction in real time, using only data observable at 14:01.

9 References

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